

Summary: [Model Methodology] LGD Methodology and Formula Documentation

Description: SMI3 uses LGD as an input to calculate expected loss and loss ratio. The current implementation also applies different HPI assumptions for the main LGD and backtesting LGD calculations.

Please provide:

- LGD calculation methodology and formula.
- LGD inputs and assumptions.
- Definition and unit of LGD output.
- HPI treatment used in LGD.
- Sample LGD calculation.

Acceptance Criteria:

- LGD methodology and formula are fully documented.
- LGD inputs, assumptions, transformations, and output units are clearly defined.

Summary: [Model Methodology] Model Selection and Hyperparameter Justification

Description: SMI3 uses AutoGluon with XGBoost as the selected model methodology. The submitted configuration restricts the model to XGBoost and does not perform hyperparameter optimization.

Please provide:

- Rationale for selecting XGBoost.
- Details of alternative models considered during development.
- Complete list of final model hyperparameters & Rationale for the selected hyperparameter values.
- Rationale for not performing hyperparameter optimization, if applicable.

Acceptance Criteria:

- XGBoost model selection is adequately justified.
- Final hyperparameters and their selection rationale are documented.
- Decision regarding hyperparameter optimization is explained.
- Selected model demonstrates acceptable performance and stability.

Summary: [Model Performance] Model Calibration Assessment

Description: Model calibration should demonstrate that predicted claim probabilities appropriately correspond to observed claim/event rates. Calibration should be assessed on development and out-of-time populations and across material portfolio segments.

Please provide:

- Predicted versus actual event/claim rates for development and out-of-time datasets.
- Calibration results by model risk decile/band.
- Calibration assessment across relevant time periods and/or regions.
- Explanation of any material overprediction or underprediction.
- Details and rationale for any calibration adjustment, scaling, floor/cap, or post-model adjustment applied.

Acceptance Criteria:

- Overall model calibration is assessed and documented.
- Predicted versus observed results are provided by risk band/decile.
- Material overprediction or underprediction is explained.
- Any calibration adjustments are documented and justified.
- Calibration results meet the model's approved performance criteria or deviations are appropriately addressed.